



CIE A Level Chemistry



Your notes

7.1 Chemical Equilibria: Reversible Reactions & Dynamic Equilibrium

Contents

- * Chemical Equilibria
- * Le Chatelier's Principle
- * Equilibrium Constants, K_c & K_p
- * Equilibrium Constant Calculations
- * Changes Affecting the Equilibrium Constant
- * Equilibria in Industrial Processes



Your notes

Chemical Equilibria

Reversible Reactions & Dynamic Equilibrium

Reversible reaction

- Some reactions go to completion where the reactants are used up to form the products and the reaction stops when all of the reactants are used up
- In **reversible reactions** the products can react to reform the original reactants
- To show a reversible reaction, two opposing half arrows are used: $\rightleftharpoons$

How reactions can be reversible

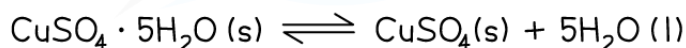
FORWARD REACTION



HYDRATED
COPPER (II)
SULFATE

ANHYDROUS
COPPER (II)
SULFATE

BACKWARD REACTION



Copyright © Save My Exams. All Rights Reserved

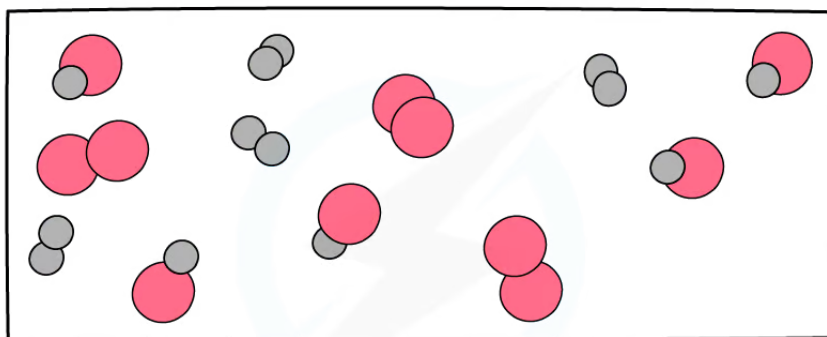
The diagram shows an example of a forward and backward reaction that can be written as one equation using two half arrows

Dynamic equilibrium

- In a **dynamic equilibrium** the reactants and products are **dynamic** (they are constantly moving)
- In a dynamic equilibrium:
 - The **rate** of the **forward** reaction is the same as the rate of the **backward** reaction in a **closed system**
 - The **concentrations** of the **reactants** and **products** are **constant**



Your notes



KEY

● = HYDROGEN ATOM ● = IODINE ATOM

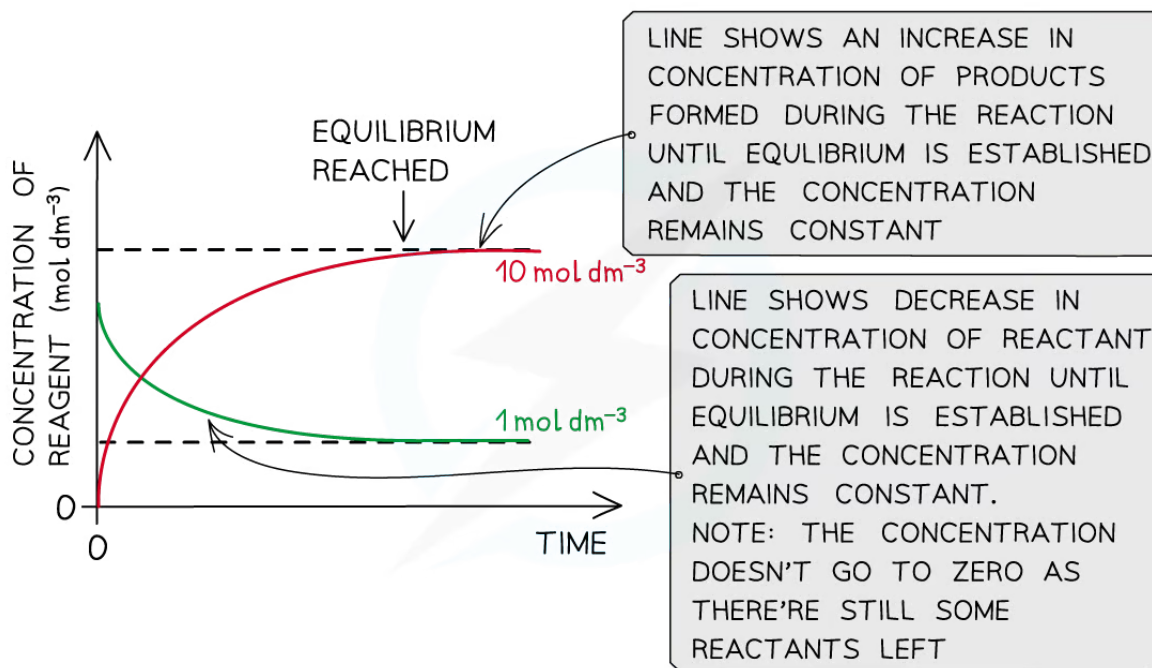
Copyright © Save My Exams. All Rights Reserved

The diagram shows a snapshot of a dynamic equilibrium in which molecules of hydrogen iodide are breaking down to hydrogen and iodine at the same rate as hydrogen and iodine molecules are reacting together to form hydrogen iodide

Diagrams showing reactant and product concentration as a reaction approaches equilibrium

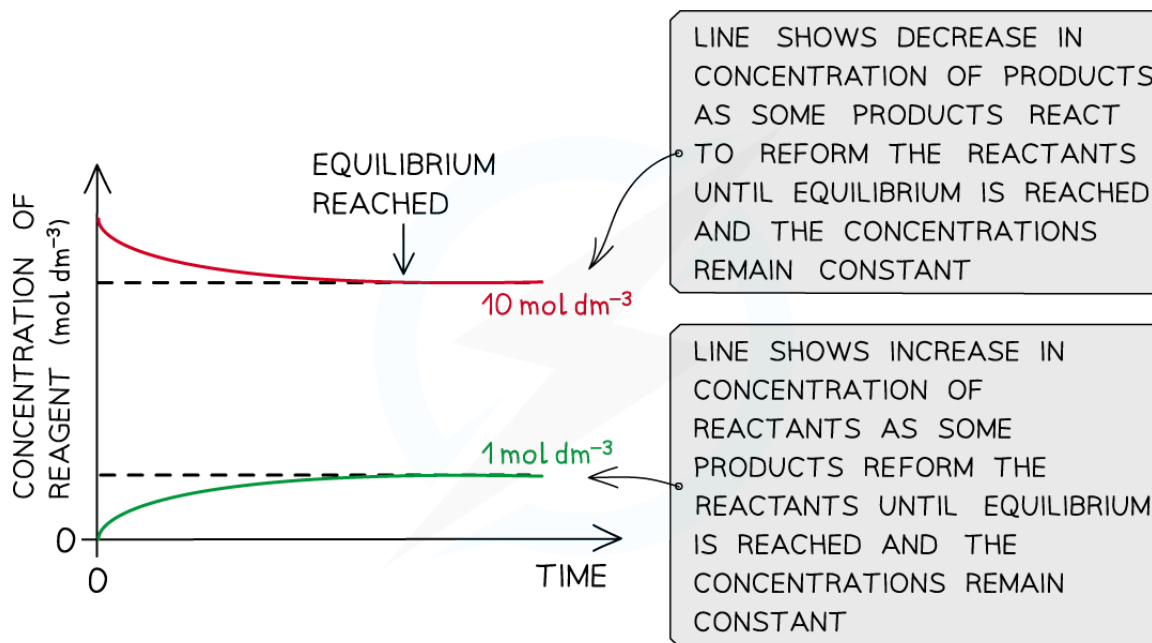


Your notes



Copyright © Save My Exams. All Rights Reserved

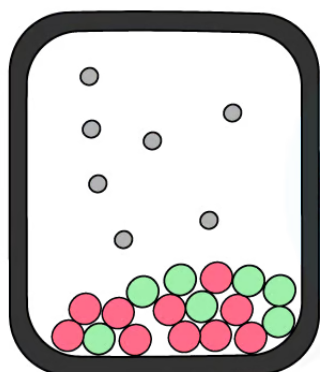
The diagram shows that the concentration of the reactants and products does not change anymore once equilibrium has been reached (equilibrium was approached using reactants)



The diagram shows that the concentration of the reactants and products does not change anymore once equilibrium has been reached (equilibrium was approached using products)

- A **closed system** is one in which none of the reactants or products escape from the reaction mixture
- In an **opensystem**, matter and energy can be lost to the surroundings
- When a reaction takes place entirely in solution, equilibrium can be reached in open flasks as a negligible amount of material is lost through evaporation
- If the reaction involves gases, equilibrium can only be reached in a closed system

Thermal decomposition of calcium carbonate in a closed system



KEY

 = $\text{CaCO}_3(\text{s})$

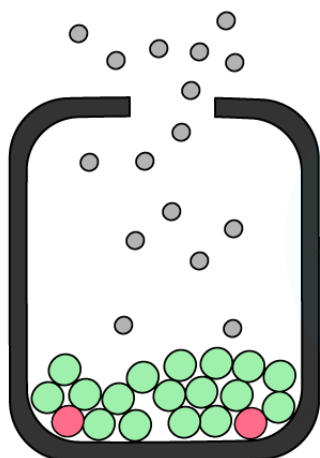
 = $\text{CaO}(\text{s})$

 = $\text{CO}_2(\text{g})$

Copyright © Save My Exams. All Rights Reserved

The diagram shows a closed system in which no carbon dioxide gas can escape and the calcium carbonate is in equilibrium with the calcium oxide and carbon dioxide

Thermal decomposition of calcium carbonate in an open system



Copyright © Save My Exams. All Rights Reserved

The diagram shows an open system in which the calcium carbonate is continually decomposing as the carbon dioxide is lost causing the reaction to eventually go to completion

 Exam Tip

A common misconception is to think that the concentrations of the reactants and products are **equal**.

However, they are **not equal** but they **remain constant** at dynamic equilibrium (i.e. the concentrations are not changing).

The concentrations will change as the reaction progresses, only **until the equilibrium** is reached.



Your notes



Your notes

Le Chatelier's Principle

Le Chatelier's Principle

Position of the equilibrium

- The **position of the equilibrium** refers to the relative amounts of products and reactants in an equilibrium mixture.
- When the position of equilibrium shifts to the **left**, it means the concentration of **reactants** increases
- When the position of equilibrium shifts to the **right**, it means the concentration of **products** increases

Le Chatelier's principle

- Le Chatelier's principle** says that if a change is made to a system at dynamic equilibrium, the position of the equilibrium moves to minimise this change
- The principle is used to predict changes to the position of equilibrium when there are changes in temperature, pressure or concentration

Effects of concentration

Effects of concentration table

Change	How the equilibrium shifts
Increase in concentration of a reactant	Equilibrium shifts to the right to reduce the effect of an increase in the concentration of a reactant
Decrease in concentration of a reactant	Equilibrium shifts to the left to reduce the effect of a decrease in the concentration of a reactant (or an increase in the concentration of product)
Increase in concentration of a product	Equilibrium shifts to the left to reduce the effect of an increase in the concentration of a product
Decrease in concentration of a product	Equilibrium shifts to the right to reduce the effect of a decrease in the concentration of a product

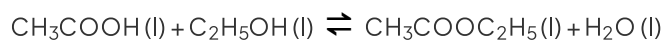


Your notes

Worked example

Changes in equilibrium position

Use the reaction below:



Explain what happens to the position of equilibrium when:

1. More $\text{CH}_3\text{COOC}_2\text{H}_5(\text{l})$ is added
2. Some $\text{C}_2\text{H}_5\text{OH}(\text{l})$ is removed

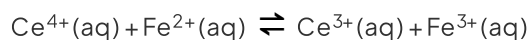
Answer

- **Answer 1:** More $\text{CH}_3\text{COOC}_2\text{H}_5(\text{l})$ is added
 - The position of the equilibrium moves **to the left** and more ethanoic acid and ethanol are formed
 - The reaction moves in this direction to oppose the effect of added ethyl ethanoate, so the ethyl ethanoate decreases in concentration
- **Answer 2:** Some $\text{C}_2\text{H}_5\text{OH}(\text{l})$ is removed
 - The position of the equilibrium moves **to the left** and more ethanoic acid and ethanol are formed
 - The reaction moves in this direction to oppose the removal of ethanol so more ethanol (and ethanoic acid) are formed from ethyl ethanoate and water

Worked example

Changes in equilibrium position

Use the reaction below:



Explain what happens to the position of equilibrium when:

1. Water is added to the equilibrium mixture.

Answer

- There is **no effect** as the water dilutes all the ions equally so there is no change in the ratio of reactants to products

Effects of pressure

- Changes in pressure only affect reactions where the reactants or products are gases

Effects of pressure table

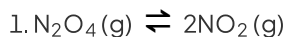
Change	How the equilibrium shifts
Increase in pressure	Equilibrium shifts in the direction that produces a smaller number of molecules of gas to decrease the pressure again
Decrease in pressure	Equilibrium shifts in the direction that produces a larger number of molecules of gas to increase the pressure again



Worked example

Changes in pressure

Predict the effect of increasing pressure on the following reactions:



Answer

- Answer 1:**
 - The equilibrium shifts to the left as there are fewer gas molecules on the left
 - This causes a decrease in pressure
- Answer 2:**
 - The equilibrium shifts to the left as there are no gas molecules on the left but there is CO_2 on the right
 - This causes a decrease in pressure



Worked example

Changes in pressure

Predict the effect of decreasing pressure on the following reaction:



Answer

- The equilibrium shifts to the right as there is a greater number of gas molecules on the right
- This causes an increase in pressure

Effects of temperature

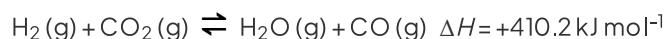
Effects of temperature table

Change	How the equilibrium shifts
Increase in temperature	Equilibrium moves in the endothermic direction to reverse the change
Decrease in temperature	Equilibrium moves in the exothermic direction to reverse the change

Worked example

Changes in temperature

Predict the effect of increasing the temperature on the following reaction:



Answer

- The reaction will absorb the excess energy
- Since the forward reaction is endothermic, the **equilibrium will shift to the right**



Your notes

 Worked example**Changes in temperature**

For the following reaction, increasing the temperature increases the amount of $\text{CO}_2(\text{g})$ at constant pressure.



Explain whether the reaction is endothermic or exothermic.

Answer

- The reaction absorbs the excess energy
- Since more $\text{CO}_2(\text{g})$ is formed, the equilibrium has shifted towards the right
- Therefore, the reaction is **endothermic**
- **Remember:** Endothermic reactions favour the products

Effects of catalysts

- A catalyst is a substance that increases the rate of a chemical reaction (they increase the rate of the **forward** and **reverse** reaction **equally**)
- Catalysts only cause a reaction to reach its equilibrium **faster**
- Catalysts therefore have **no effect** on the **position of the equilibrium** once this is reached



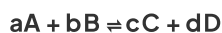
Your notes

Equilibrium Constants, K_c & K_p

Equilibrium Constant: Concentrations

Equilibrium expression & constant

- The **equilibrium expression** is an expression that links the **equilibrium constant**, K_c , to the **concentrations of reactants and products** at equilibrium taking the **stoichiometry** of the equation into account
- So, for a given reaction:



- K_c is defined as:

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

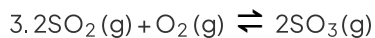
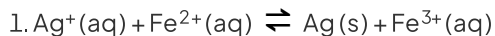
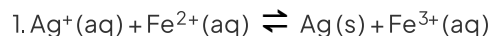
- Where:
 - [A] and [B] are the equilibrium concentrations of **A** and **B**, in mol dm^{-3}
 - [C] and [D] are the equilibrium concentrations of **C** and **D**, in mol dm^{-3}
 - a, b, c and d are the respective number of moles of each reactant and product
- Solids** are ignored in equilibrium expressions
- The K_c of a reaction is specific and only changes if the **temperature** of the reaction changes



Your notes

 Worked example

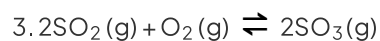
Deduce the equilibrium expression for the following reactions:

**Answer**

$$K_c = \frac{[\text{Fe}^{3+}(\text{aq})]}{[\text{Fe}^{2+}(\text{aq})][\text{Ag}^+(\text{aq})]}$$



$$K_c = \frac{[\text{NH}_3(\text{g})]^2}{[\text{N}_2(\text{g})][\text{H}_2(\text{g})]^3}$$



$$K_c = \frac{[\text{SO}_3(\text{g})]^2}{[\text{SO}_2(\text{g})]^2[\text{O}_2(\text{g})]}$$



Your notes

Mole Fraction & Partial Pressure

Partial pressure

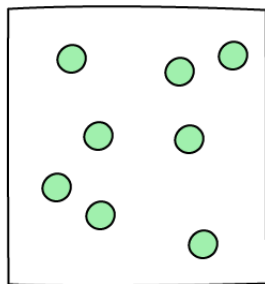
- For reactions involving mixtures of gases, the equilibrium constant K_p is used as it is easier to measure the **pressure** than the concentration for gases
- The **partial pressure** of a gas is the pressure that the gas would have if it was in the container all by itself
- The **total pressure** is the sum of the **partial pressures**:

$$P_{\text{total}} = P_A + P_B + P_C + \dots$$

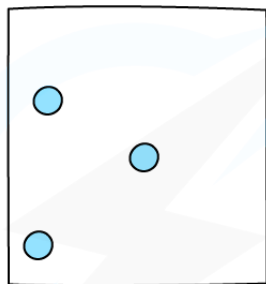
- P_{total} = total pressure
- P_A, P_B, P_C = partial pressures

How partial pressures contribute to total pressure

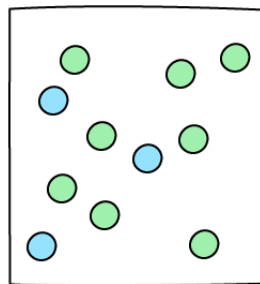
THE VOLUMES OF THE CONTAINERS ARE THE SAME



PARTIAL PRESSURE
= 8 Pa



PARTIAL PRESSURE
= 3 Pa



TOTAL PRESSURE
= 11 Pa

KEY

 = GAS A

 = GAS B

Copyright © Save My Exams. All Rights Reserved

Partial pressures can be added together to calculate the total pressure

Mole fraction

- The mole fraction of a gas is the ratio of moles of a particular gas to the total number of moles of gas present

$$\text{Mole fraction} = \frac{\text{number of moles of a particular gas}}{\text{total number of moles of all the gases in the mixture}}$$

- To calculate the **partial pressures of each gas** the following relationship can be used:

$$\text{Partial pressure} = \text{mole fraction} \times \text{total pressure}$$

- The sum of the mole fractions should add up to 1.00, while the sum of the partial pressures should add up to the total pressure



Your notes

Equilibrium Constant: Partial Pressures

Equilibrium expressions involving partial pressures

- Equilibrium expressions in terms of partial pressures are written similarly to those involving concentrations with a few differences:

Comparing K_p and K_c expressions

$$K_p = \frac{P_C^c \times P_D^d}{P_A^a \times P_B^b}$$

EXPRESSION CONSTANT

PARTIAL PRESSURE

NUMBER OF MOLES OF
REACTANTS AND PRODUCTS

REACTANTS AND PRODUCTS

$$K_c = \frac{[C]^c \times [D]^d}{[A]^a \times [B]^b}$$

EXPRESSION CONSTANT

BRACKETS USED FOR
CONCENTRATION

NUMBER OF MOLES OF
REACTANTS AND PRODUCTS

REACTANTS AND PRODUCTS

Copyright © Save My Exams. All Rights Reserved

The process of writing the expressions is similar, but there is a different presentation and different information required



Your notes

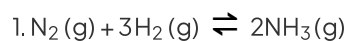


Your notes

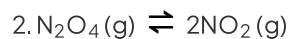
 Worked example**Deducing equilibrium expressions of gaseous reactions**

Deduce the equilibrium expression for the following reactions:

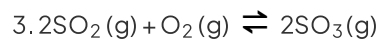
1. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
2. $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
3. $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$

Answer

$$K_p = \frac{p^2 \text{NH}_3}{p^3 \text{H}_2 \times p \text{N}_2}$$



$$K_p = \frac{p^2 \text{NO}_2}{p \text{N}_2\text{O}_4}$$



$$K_p = \frac{p^2 \text{SO}_3}{p^2 \text{SO}_2 \times p \text{O}_2}$$



Your notes

Equilibrium Constant Calculations

Equilibrium Constant: Calculations

Calculations involving K_c

- In the equilibrium expression, each figure within a square bracket represents the concentration in **mol dm^{-3}**
- The **units** of K_c therefore depend on the form of the equilibrium expression
- Some questions give the **number of moles** of each of the reactants and products at equilibrium together with the volume of the reaction mixture
- The concentrations of the reactants and products can then be calculated from the number of moles and total volume using:

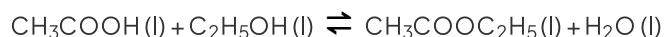
$$\text{concentration } (\text{mol dm}^{-3}) = \frac{\text{number of moles}}{\text{volume } (\text{dm}^3)}$$



Your notes

Worked example

At equilibrium, 500 cm³ of the following reaction mixture contains 0.235 mol of ethanoic acid, 0.0350 mol of ethanol, 0.182 mol of ethyl ethanoate and 0.182 mol of water.



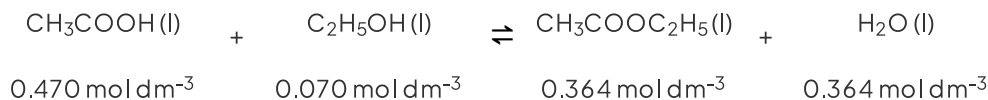
Use this information to calculate a value of K_c for this reaction.

Answer

- **Step 1:** Calculate the concentrations of the reactants and products:

- $[\text{CH}_3\text{COOH}(\text{l})] = \frac{0.235}{0.500} = 0.470 \text{ mol dm}^{-3}$
- $[\text{C}_2\text{H}_5\text{OH}(\text{l})] = \frac{0.0350}{0.500} = 0.070 \text{ mol dm}^{-3}$
- $[\text{CH}_3\text{COOC}_2\text{H}_5(\text{l})] = \frac{0.182}{0.500} = 0.364 \text{ mol dm}^{-3}$
- $[\text{H}_2\text{O}(\text{l})] = \frac{0.182}{0.500} = 0.364 \text{ mol dm}^{-3}$

- **Step 2:** Write out the balanced chemical equation with the calculated concentrations beneath each substance:



- **Step 3:** Write the equilibrium constant for this reaction in terms of concentration:

- $K_c = \frac{[\text{H}_2\text{O}][\text{CH}_3\text{COOC}_2\text{H}_5]}{[\text{C}_2\text{H}_5\text{OH}][\text{CH}_3\text{COOH}]}$

- **Step 4:** Substitute the equilibrium concentrations into the expression:

- $K_c = \frac{0.364 \times 0.364}{0.070 \times 0.470}$
- $K_c = 4.03$

- **Step 5:** Deduce the correct units for K_c :

- $K_c = \frac{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}$
- All units cancel out
- Therefore, $K_c = 4.03$

- Note that the smallest number of significant figures used in the question is 3, so the final answer should also be given to 3 significant figures

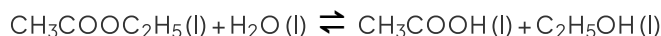
- Some questions give the **initial and equilibrium concentrations** of the reactants but products
- An initial, change and equilibrium table should be used to determine the equilibrium concentration of the products **using the molar ratio of reactants and products in the stoichiometric equation**



Your notes

Worked example

Ethyl ethanoate is hydrolysed in water.



0.1000 mol of ethyl ethanoate are added to 0.1000 mol of water. A little acid catalyst is added and the mixture is made up to 1 dm³.

At equilibrium, 0.0654 mol of water are present.

Use this information to calculate a value of K_c for this reaction.

Answer

- Step 1:** Write the balanced chemical equation, with the concentrations beneath each substance, into an initial, change and equilibrium (ICE) table:

	$\text{CH}_3\text{COOC}_2\text{H}_5(\text{l})$	$+\text{H}_2\text{O}(\text{l})$	$\rightleftharpoons$	$\text{CH}_3\text{COOH}(\text{l})$	$+\text{C}_2\text{H}_5\text{OH}(\text{l})$
Initial moles	0.1000	0.1000		0	0
Change	-0.0346	-0.0346		+0.0346	+0.0346
Equilibrium moles	0.0654	0.0654		0.0346	0.0346

- Step 2:** Calculate the concentrations of the reactants and products:

- $[\text{H}_2\text{O}(\text{l})] = \frac{0.0654}{1.000} = 0.0654 \text{ mol dm}^{-3}$
- $[\text{CH}_3\text{COOC}_2\text{H}_5(\text{l})] = \frac{0.0654}{1.000} = 0.0654 \text{ mol dm}^{-3}$
- $[\text{C}_2\text{H}_5\text{OH}(\text{l})] = \frac{0.0346}{1.000} = 0.0346 \text{ mol dm}^{-3}$
- $[\text{CH}_3\text{COOH}(\text{l})] = \frac{0.0346}{1.000} = 0.0346 \text{ mol dm}^{-3}$

- Step 3:** Write the equilibrium constant for this reaction in terms of concentration:

$$K_c = \frac{[\text{C}_2\text{H}_5\text{OH}][\text{CH}_3\text{COOH}]}{[\text{H}_2\text{O}][\text{CH}_3\text{COOC}_2\text{H}_5]}$$



Your notes

- **Step 4:** Substitute the equilibrium concentrations into the expression:

- $K_c = \frac{0.0346 \times 0.0346}{0.0654 \times 0.0654}$

- $K_c = 0.28$

- **Step 5:** Deduce the correct units for K_c :

- $K_c =$

- All units cancel out

- Therefore, $K_c = 0.28$

Calculations involving K_p

- In the equilibrium expression the p represent the partial pressure of the reactants and products in Pa
- The **units** of K_p therefore depend on the form of the equilibrium expression



Your notes

Worked example

The equilibrium between sulfur dioxide, oxygen and sulfur trioxide is as follows:



At constant temperature, the equilibrium partial pressures are:

- $\text{SO}_2 = 1.0 \times 10^6 \text{ Pa}$
- $\text{O}_2 = 7.0 \times 10^6 \text{ Pa}$
- $\text{SO}_3 = 8.0 \times 10^6 \text{ Pa}$

Calculate the value of K_p for this reaction.

Answer

- **Step 1:** Write the equilibrium constant for the reaction in terms of partial pressures:

$$K_p = \frac{p^2 \text{SO}_3}{p^2 \text{SO}_2 \times p \text{O}_2}$$

- **Step 2:** Substitute the equilibrium concentrations into the expression:

$$K_p = \frac{(8.0 \times 10^6)^2}{(1.0 \times 10^6)^2 \times (7.0 \times 10^6)}$$

$$K_p = 9.1 \times 10^{-6}$$

- **Step 3:** Deduce the correct units of K_p :

$$K_p = \frac{\text{Pa}^2}{\text{Pa}^2 \times \text{Pa}}$$

So, the units of K_p are Pa^{-1}

Therefore, $K_p = 9.1 \times 10^{-6} \text{ Pa}^{-1}$

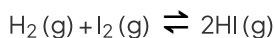
- Some questions only give the **number of moles** of gases present and the total pressure
- The number of moles of each gas should be used to first calculate the **mole fractions**
- The mole fractions are then used to calculate the **partial pressures**
- The values of the partial pressures are then substituted in the **equilibrium expression**



Your notes

Worked example

The equilibrium between sulfur dioxide, oxygen and sulfur trioxide is as follows:



At constant temperature, the equilibrium partial pressures are:

- $\text{H}_2 = 1.71 \times 10^{-3} \text{ Pa}$
- $\text{I}_2 = 2.91 \times 10^{-3} \text{ Pa}$
- $\text{HI} = 1.65 \times 10^{-3} \text{ Pa}$

The total pressure is 100 kPa.

Calculate the value of K_p for this reaction.

Answer

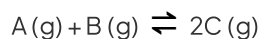
- **Step 1:** Calculate the total number of moles:
 - Total number of moles = $1.71 \times 10^{-3} + 2.91 \times 10^{-3} + 1.65 \times 10^{-2}$
 - Total number of moles = 2.112×10^{-2}
- **Step 2:** Calculate the mole fraction of each gas:
 - $\text{H}_2 = \frac{1.71 \times 10^{-3}}{2.112 \times 10^{-2}} = 0.0810$
 - $\text{I}_2 = \frac{2.91 \times 10^{-3}}{2.112 \times 10^{-2}} = 0.1378$
 - $\text{HI} = \frac{1.65 \times 10^{-2}}{2.112 \times 10^{-2}} = 0.7813$
- **Step 3:** Calculate the partial pressure of each gas:
 - $\text{H}_2 = 0.0810 \times 100 = 8.10 \text{ kPa}$
 - $\text{I}_2 = 0.1378 \times 100 = 13.78 \text{ kPa}$
 - $\text{HI} = 0.7813 \times 100 = 78.13 \text{ kPa}$
- **Step 4:** Write the equilibrium constant in terms of partial pressure:
 - $K_p = \frac{p^2 \text{ HI}}{p \text{ H}_2 \times p \text{ I}_2}$
- **Step 5:** Substitute the values into the equilibrium expression:
 - $K_p = \frac{78.13^2}{8.10 \times 13.78}$
 - $K_p = 54.7$
- **Step 6:** Deduce the correct units for K_p :
 - $K_p = \frac{\text{Pa}^2}{\text{Pa} \times \text{Pa}}$



- All units cancel out
 - Therefore, $K_p = 54.7$
-
- Other questions related to equilibrium expressions may involve calculating quantities present at equilibrium given appropriate data

Worked example

An equilibrium is set up in a closed container between equal volumes of gaseous reactants **A** and **B** to form a gaseous product **C**.



The total pressure within the container, at 50 °C, is 3 atm.

The equilibrium partial pressure of **A**, at 50 °C, is 0.5 atm.

What is the equilibrium partial pressure of **C** at this temperature?

Answer

- There are equal volumes of reactants **A** and **B** in a 1:1 molar ratio
 - This means their partial pressures will be the same.
 - **B** therefore also has an equilibrium partial pressure of 0.5 atm
- **Total pressure = Σ (equilibrium partial pressures)**
 - Therefore, the sum of all the partial pressures must equal to 3 atm
 - $0.5 + 0.5 + p_{\text{C}} = 3 \text{ atm}$
 - $p_{\text{C}} = 2 \text{ atm}$



Your notes

Changes Affecting the Equilibrium Constant

Changes that Affect the Equilibrium Constant

Changes in concentration

- If all other conditions stay the same, the equilibrium constant K_c is **not affected** by any changes in concentration of the reactants or products

- For example, the decomposition of hydrogen iodide:



- The equilibrium expression is:

$$K_c = \frac{[\text{H}_2][\text{I}_2]}{[\text{HI}]^2} = 6.25 \times 10^{-3}$$

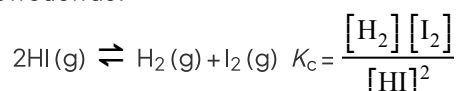
- Adding more HI makes the ratio of [products] to [reactants] smaller
- To restore equilibrium, $[\text{H}_2]$ and $[\text{I}_2]$ increase and $[\text{HI}]$ decreases
- Equilibrium is restored when the ratio is 6.25×10^{-3} again

Changes in pressure

- A change in pressure **only** changes the **position of the equilibrium** (see [Le Chatelier's principle](#))
- If all other conditions stay the same, the equilibrium constant K_c is **not affected** by any changes in the pressure of the reactants and products

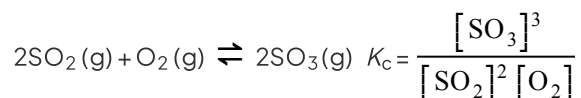
Changes in temperature

- Changes in temperature **change** the equilibrium constant K_c
- For an endothermic reaction such as:



- With an increase in temperature:
 - $[\text{H}_2]$ and $[\text{I}_2]$ **increases**
 - $[\text{HI}]$ **decreases**
 - Because $[\text{H}_2]$ and $[\text{I}_2]$ **increase** and $[\text{HI}]$ **decreases**, the equilibrium constant K_c **increases**

- For an exothermic reaction such as:



- With an increase in temperature:



Your notes

- $[\text{SO}_3]$ **decreases**
- $[\text{SO}_2]$ and $[\text{O}_2]$ **increases**
- Because $[\text{SO}_3]$ **decreases** and $[\text{SO}_2]$ and $[\text{O}_2]$ **increase**, the equilibrium constant K_c **decreases**

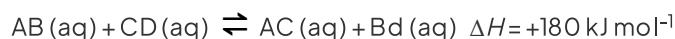
Presence of a catalyst

- If all other conditions stay the same, the equilibrium constant K_c is **not affected** by the presence of a catalyst
- A catalyst speeds up both the forward and reverse reactions at the same rate so the ratio of [products] to [reactants] remains unchanged

Worked example

Factors affecting K_c

An equilibrium is established in the following reaction:



Which factors would affect the value of K_c in this equilibrium?

Answer

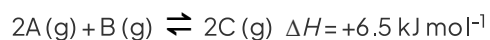
- Only a change in temperature will affect the value of K_c
- Any other changes in conditions would result in the position of the equilibrium moving to oppose this change
- Adding a catalyst increases the rate of reaction meaning the state of equilibrium will be reached faster but has no effect on the position of the equilibrium and, therefore, K_c is unchanged



Your notes

 Worked example**Factors which increase K_p value**

What will increase the value of K_p for the following equilibrium?

**Answer**

- Only temperature changes permanently affect the value of K_p
- An increase in temperature shifts the reaction in favour of the products
- The [products] increases and [reactants] decreases, therefore, the K_p value increases



Your notes

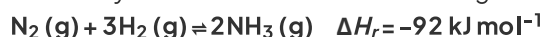
Equilibria in Industrial Processes

Haber & Contact Processes

- Equilibrium reactions are involved in some stages of large-scale production of certain chemicals
- An understanding of equilibrium and Le Chatelier's principle is therefore very important in the chemical industry

Haber process

- The Haber process involves the synthesis of ammonia according to:



- Le Chatelier's principle is used to get the best yield of ammonia

Maximising the ammonia yield

Pressure

- An increase in **pressure** will result in the equilibrium shifting in the direction of the fewest molecules of gas formed to reduce the pressure
- In this case, the equilibrium shifts towards the right so the yield of ammonia increases
- An increase in pressure will cause the particles to be closer together and therefore increasing the number of successful collisions leading to an increased reaction rate
- Very high pressures are expensive to produce therefore a **compromise pressure** of 200 atm is chosen

Temperature

- To get the maximum yield of ammonia the position of equilibrium should be shifted as far as possible to the right as possible
- Since the Haber process is an exothermic reaction, according to Le Chatelier's principle the equilibrium will shift to the right if the temperature is lowered
- A decrease in **temperature** will decrease the energy of the surroundings so the reaction will go in the direction in which energy is released to counteract this
- Since the reaction is exothermic, the equilibrium shifts to the right
- **However**, at a low temperature the gases won't have enough kinetic energy to collide and react and therefore equilibrium would not be reached therefore **compromise temperature** of 400–450 °C is used in the Haber process
- A **heat exchanger** warms the incoming gas mixture to give molecules more **kinetic energy** such that the gas molecules **collide more frequently** increasing the likelihood of a reaction

Removing ammonia



Your notes

- **Removing ammonia by condensing it to a liquid** causes the equilibrium position to shift to the right to replace the ammonia causing more ammonia to be formed from hydrogen and nitrogen
- The removed ammonia is stored at **very low temperatures** and there is **no catalyst** present with the stored ammonia so the decomposition reaction of ammonia to decompose back into hydrogen and nitrogen will be too slow

Catalysts

- In the absence of a catalyst the reaction is so slow that hardly anything happens in a reasonable time!
- Adding an iron catalyst speeds up the rate of reaction

Contact process

- The Contact process involves the synthesis of sulfuric acid according to:



- Le Chatelier's principle is used to get the best yield of sulfuric acid

Maximising the sulfuric acid yield

Pressure

- An increase in **pressure** will result in the equilibrium shifting in the direction of the fewest molecules of gas formed to reduce the pressure
- In this case, the equilibrium shifts towards the right so the yield of sulfuric acid increases
- In practice, the reaction is carried out at only 1 atm
- This is because K_p for this reaction is already very high meaning that the position of the equilibrium is already far over to the right
- Higher pressures than 1 atm will be unnecessary and expensive

Temperature

- The same principle applies to increasing the temperature in the Contact process as in the Haber process
- A **compromise temperature** of 450 °C is used

Removing sulfuric acid

- SO_3 is removed by absorbing it in 98% sulfuric acid
- The SO_3 reacts with the solution and more H_2SO_4 is formed

Catalysts

- The Contact process uses vanadium(V) oxide as a catalyst to increase the rate of reaction